MEC 101 Robot Design Project: Team 5 “The Purwarriors”

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YouTube Link: https://youtube.com/shorts/zyL90W8pS90?si=U88hbHphQnXdbIb9

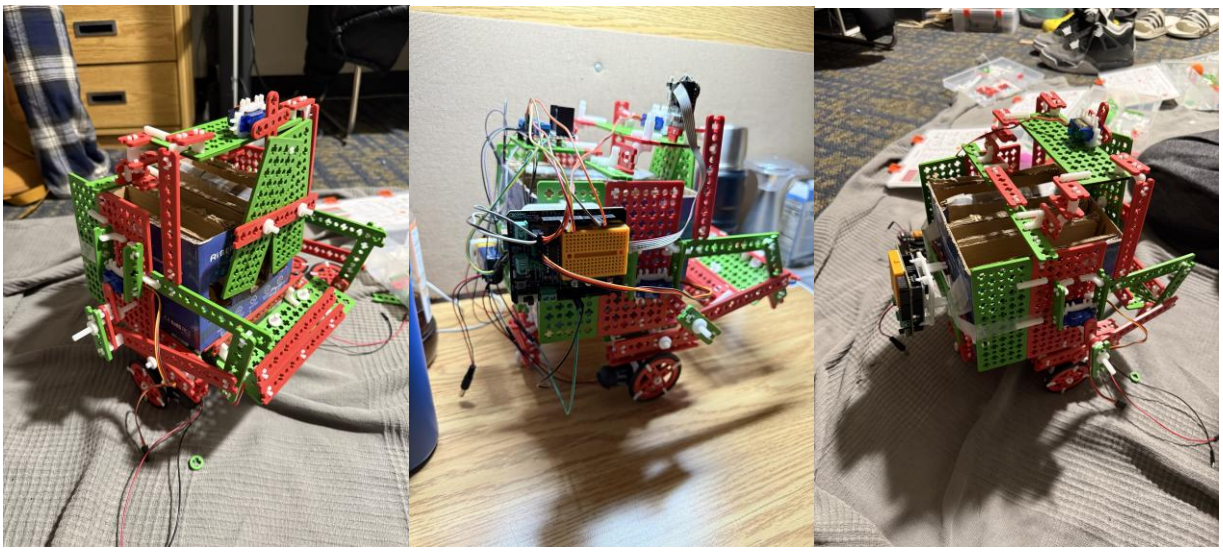
(We couldn’t make a video because YouTube won't allow it unless it’s over 3 minutes)

Introduction/Description:

Our team's robot is a candy dispensing robot which is remotely operated by a combination of sensors and linkage bars which will allow it to present candies (more specifically Peanut M&Ms) to a nearby observer. There are two ways to jumpstart the robot's movements: object recognition and color recognition.

Beginning with object recognition, the robot is programmed to sense the presence of a purple pin and a yellow notebook. The reason why we chose these objects is because they have bright solid colors or a definable shape, which makes it easier for the Pixy camera sensor mounted on top of the robot to detect what is in front of it. Verification between the objects is different however: the notebook's job is to allow the robot to proceed in dispensing the M&Ms. The recognition of the notebook also wakes up the four-bar linkage system and the movement of the wheels which prepare for handing the candy to the observer. On the other hand, the sharpie makes the robot suspend all dispensing actions beyond recognition. This can be useful for executing a mid-stop or cancellation.

Color recognition is the second factor in making the robot function. In the demonstration, we used three different colored sheets to resemble the three different M&M candy colors coming from the three individual chutes. To activate this, place a color sheet in front of the robot's Pixy camera; the robot will detect the specific color and move its doors to allow that color M&M to flow out and onto the platform. For example, if we want the robot to dispense a red M&M, we would show a red piece of colored paper in front of the robot's pixy camera.



Figures 1,2,3: The candy dispensing robot in its completed state; the front, middle, and rear proportions of the robot are shown above.

Mechanism:

There are three main mechanisms that provide the movement of our robot: Three individual slanted chutes for candies, two four-bar mechanisms to support the candy platform, and an oscillating door mechanism to permit and restrict candy flow from the chutes.

The chute sets can be described as the keystone piece for the entire robot as without them, every other component wouldn't have their purpose. We decided to use three chutes because we wanted our robot to store three different colors of M&Ms which will be released when detected. Unlike most of the other pieces which used mainly Snappy XO parts to create this robot, the chutes were created purely from cut up rectangular cardboard pieces slanted at around a 45-degree angle to allow an inclined plane ramp. To make each individual chute, a measurement had to be taken by cutting a little over the length of a cardboard box which would store the chutes internally and act as the monocoque of the entire robot. Giving each chute the extra length would thus allow for that inclined plane structure. To ensure that each individual chute stayed in place during testing, super glue was applied to the interior of the cardboard box to promote rigidity. Additionally, the widths of the chutes were also cut to a specific amount, so it could fit through three respective holes from the cardboard box (this is where the M&Ms from the chute would flow out). Inside the cardboard box, each chute would also have cardboard walls to prevent any M&Ms from spilling over. These parts were also super glued together with the chutes.

The four-bar linkage mechanism is what allows the M&M's to be presented when it exits one of the chutes. There are two systems on the robot (one for each side) which were developed through MotionGen to create an upwards-forward motion. The arms were pieced together through a set of servo motors, Snappy XO beams, clips, and circular spacers. When built, the main appearance of the arms consisted of two long and two short Snappy XO beams. These beans provided the length of the entire mechanism to stretch out and allow for the creation of a candy platform later on. To connect the beams together, each beam was placed in a large-small-large orientation to create a rectangular shape. They were then joined together by small H-Clip pivots which allowed the parts to osculate freely giving that four-bar movement. This entire system, when completed, is then connected to a servo mounted on the side of the robot's cardboard monocoque. We additionally made sure to add cylindrical tubes and spacers to allow the beams to not interfere with each other when not moving. Observing at the bottom left of the arm, there is an extension of another small beam overlapping an already joined beam. This connection allowed for not only a more secure movement of the beams, but also an easier manual alternative to move the beams if the servo powering the mechanism fails or becomes stuck. Attached between the two arms is the candy platform. Formed using two 4x Snappy XO plates, the platform moves in an outwards up and down direction thanks to the motion of the four-bar linkage system. To prevent a constant motion as the platform moves up and down, H-Clip pivots are added to the arms of each arm. There are also multiple L-clipped Snappy XO beams attached on the top and bottom of the platform to ensure that the M&Ms fall into the divot the beams create and not elsewhere.

The oscillating door of the robot sits in front of the exiting holes of the candy chutes. Powering the door is another servo motor placed on top of the robot. The servo is programmed through Arduino to a specific angle of measurement to prevent the doors from hitting the linkage arms on the side. The door covers two out of the three chute holes using two 4x Snappy XO plates connected in an intersecting direction by a Snappy XO beam in the middle. When in motion, the door will specifically move to the position it needs to which is the area where the chute holes are. This allows for the requested M&M color to come out of the cardboard. Because we did not have the time to create a custom piece, the doors are not the most elegant when looking at the rest of the robot. The servo motion additionally prevents the open chute hole from being fully exposed. Still, the M&Ms are still able to roll out just fine.

Additional Mechanics:

Following from the main mechanics, our robot also has a wheel system placed below the monocoque. The wheels allow the robot to move whenever it detects a nearby observer.

Each wheel is powered by a DC motor which is then linked to an L298 Motor Director for movement sensing.



Figure 4: 4-bar linkage system on one of two sides of the robot.

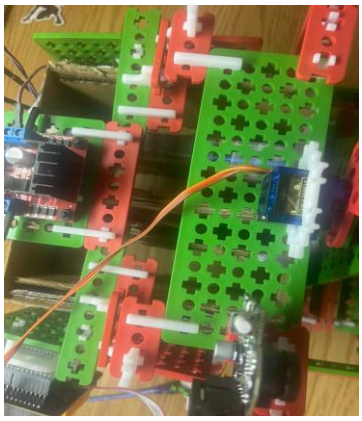


Figure 5: Upwards view of the 3-candy chute system (they are the cardboard below the structure).

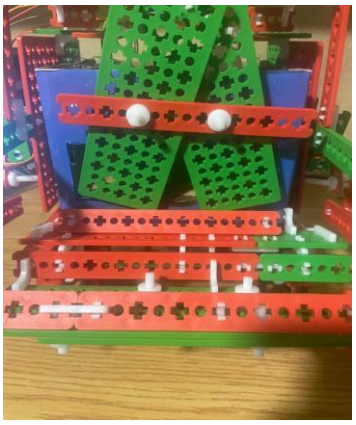


Figure 6: Frontwards view of the oscillating doors.

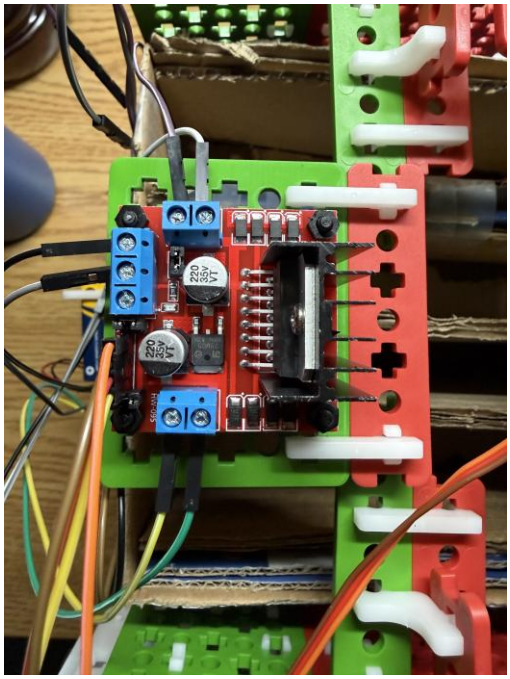


Figure 7: Closeup of the L298 Motor

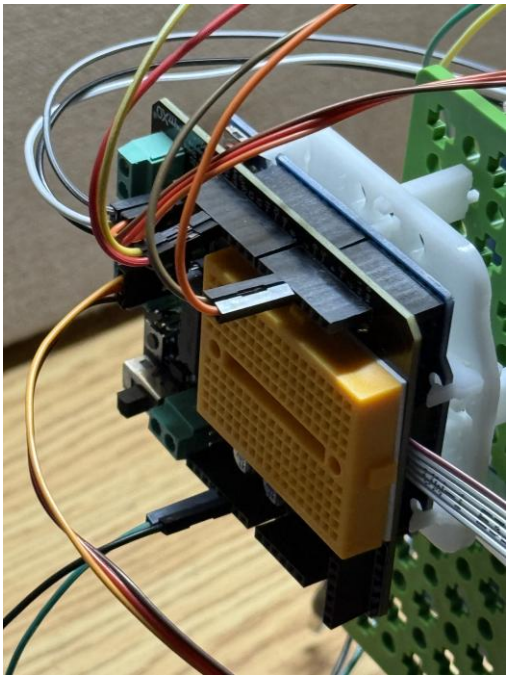


Figure 8: Closeup of the Motor Sheild and

Director which powers the robot's Wheels.

the wiring which programs the mechanisms of the robot.



Figure 9: Closeup of the Pixy sensor and the servo motor for the door.

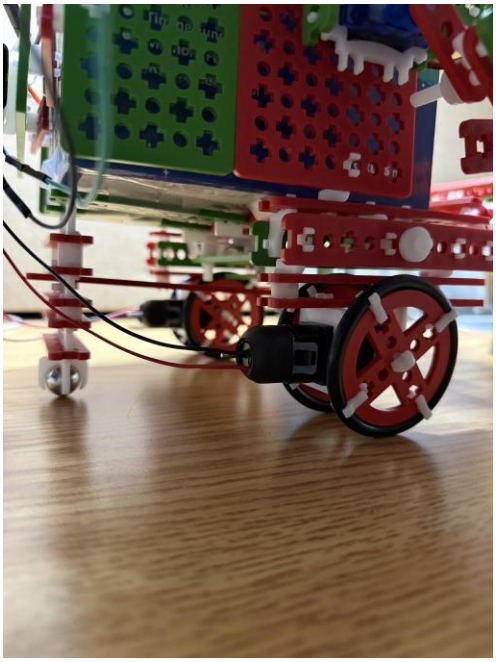


Figure 10: Closeup of the wheels and DC Motors powering them.

Custom Parts Design & Design Sketches:

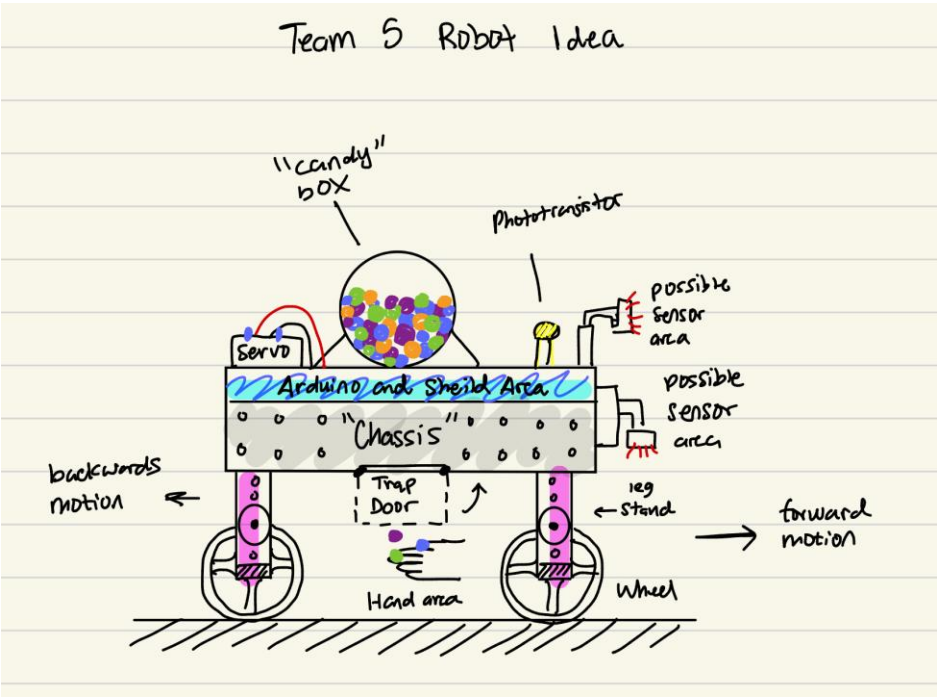


Figure 11: Original prototype sketch of candy dispenser robot which was supposed to be on four wheels rather than two and had an underside opening door.

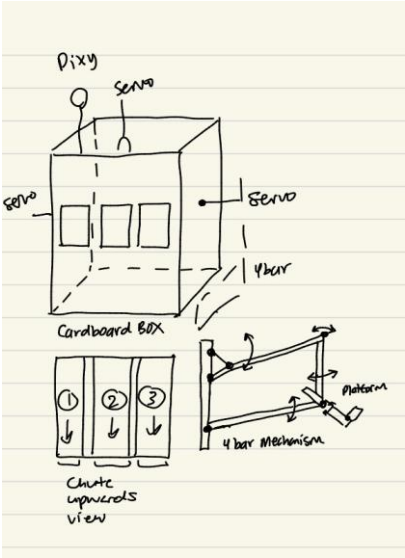


Figure 12: Second prototype sketch of candy dispenser robot; includes the four bar linkage system and chutes.

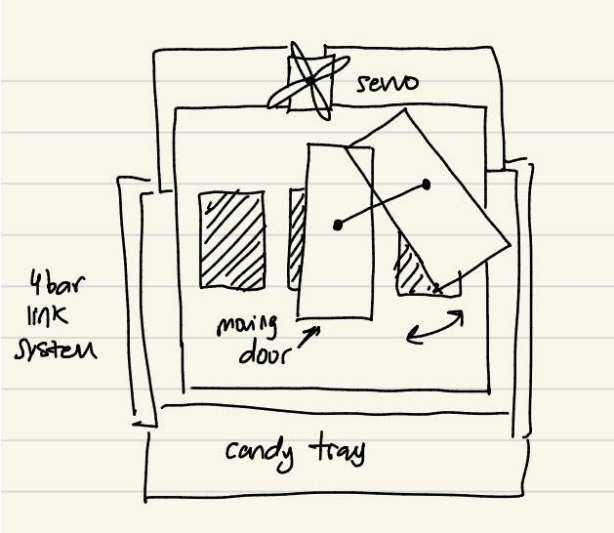


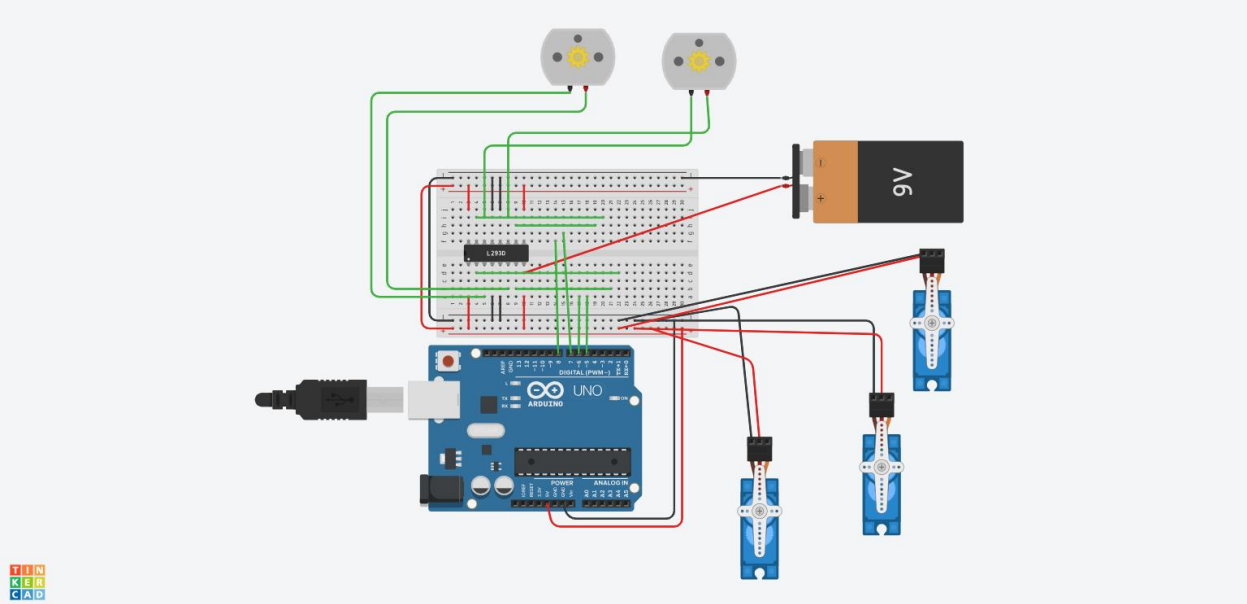
Figure 13: Sketch of servo-door mechanism.

Bill of Materials:

Quantity	Source	Description
8	Snappy XO	3x Plate
2	Snappy XO	4x Plate
6	Snappy XO	2x Beam
3	Snappy XO	3x Beam
8	Snappy XO	4x Beam
8	Snappy XO	5x Beam
8	Snappy XO	6x Beam
7	Snappy XO	7x Beam
7	Snappy XO	8x Beam
22	Snappy XO	Small H-Clip
19	Snappy XO	Medium H-Clip
4	Snappy XO	Large H-Clip
3	Snappy XO	Medium H-Clip Pivot
32	Snappy XO	L-Clip
1	Snappy XO	Metal Ball & Holder

16	Snappy XO	Wheel Clip
4	Snappy XO	Wheels & Tires
4	Snappy XO	Motor Spacer
19	Snappy XO	Circle Spacer
2	Snappy XO	DC Motor
2	Snappy XO	DC Motor Mount
3	Snappy XO	Servo Motor
3	Snappy XO	Servo Motor Mount
4	Snappy XO	Servo Motor Full Mount
1	Snappy XO	Servo Motor Plus Mount
1	Snappy XO	Arduino Uno Board
1	Snappy XO	Arduino Uno Mount
1	Snappy XO	Motor Shield
1	Snappy XO	9V Battery
30+	Snappy XO	Wires
1	HE 207 Office	Pixy Camera Sensor
1	HE 207 Office	Pixy Stand
1	HE 207 Office	L298 Motor Director
1	HE 207 Office	L298 Motor Mount
10	Recycling Bin	Cardboard
4 Tubes	Amazon	Super Glue
2	Walgreens	Scotch Tape
4 Bags	Walgreens	M&Ms Candy

Circuit Images & Arduino Code:




```
#include <Servo.h>
#include <Pixy2.h>
Pixy2 pixy;
Servo doorServo;      // door that selects chute

Servo rightServo;     // four-bar linkage
Servo leftServo;
int CHUTE_RED    = 0;

int CHUTE_GREEN  = 25;
int CHUTE_BLUE   = 40;
int CHUTE_CLOSED = 10;
int LEFTLINKAGE_CATCH = 120;
int RIGHTLINKAGE_CATCH = 20;
int LEFTLINKAGE_PRESENT = 20;

int RIGHTLINKAGE_PRESENT = 120;


void setup() {

  pixy.init();
  Serial.begin(9600);
  rightServo.attach(8);

  leftServo.attach(9);
  doorServo.attach(10);
}
void loop() {

  rightServo.write(RIGHTLINKAGE_CATCH);
  leftServo.write(LEFTLINKAGE_CATCH);
  doorServo.write(CHUTE_CLOSED);
```

```
  int blocks = pixy.ccc.getBlocks();
  if (blocks > 0) {
    for (int i = 0; i < blocks; i++) {


 int signature = pixy.ccc.blocks[i].m_signature;

      Serial.print("Detected signature: ");
      Serial.println(signature);



      // Signature 5 = turn direction
      if (signature == 5) {
        Serial.println("Signature 5 detected → Turning");
       // TurnRight();
       doorServo.write(12);
        delay(700);
         doorServo.write(8);
        delay(700);
         doorServo.write(12);
        delay(700);
        return;   // restart loop
      }
if (signature == 4) {

        Serial.println("Signature 4 detected → Stopping & checking candy color");
          //pause();

        //delay(300);
       checkCandyColor();

        return;   // after dispensing, restart loop
      }
    }
  }
}
void checkCandyColor() {

 delay(200);

  pixy.ccc.getBlocks();
 for (int i = 0; i < pixy.ccc.numBlocks; i++) {

 int s = pixy.ccc.blocks[i].m_signature;

 if (s == 1) {
      Serial.println("→ RED candy");
      doorServo.write(CHUTE_RED);
      delay(1000);
```

```
      doorServo.write(CHUTE_CLOSED);
        delay(1000);
      presentCandy();
      return;
} else if (s == 2) {
      Serial.println("→ GREEN candy");
      doorServo.write(CHUTE_GREEN);
      delay(1000);
      doorServo.write(CHUTE_CLOSED);
        delay(1000);
      presentCandy();
      return;
} else if (s == 3) {

      Serial.println("→ BLUE candy");
      doorServo.write(CHUTE_BLUE);
      delay(1000);
      doorServo.write(CHUTE_CLOSED);
      delay(1000);
      presentCandy();
      return;
    }
  }


Serial.println("No valid candy color detected.");

}
void presentCandy() {

 Serial.println("Presenting candy...");

 // extend linkage

  rightServo.write(RIGHTLINKAGE_PRESENT);
  leftServo.write(LEFTLINKAGE_PRESENT);
  delay(8000);      // hold for 8 seconds
// retract

  rightServo.write(RIGHTLINKAGE_CATCH);
  leftServo.write(LEFTLINKAGE_CATCH);
  delay(800);


  Serial.println("Candy presented → resetting.");
}
```